



Indigenously Developed AI Based Parental Control Router

KidSafeNet – Protecting Children, Shaping a Safer Digital Future.



KidSafeNet



ABSTRACT

KidSafeNet is an indigenously developed AI-based parental control router designed to enable a safer online environment for children. The system intelligently analyzes internet traffic, detects inappropriate and addictive content, blocks inappropriate content, and redirects users to safe content. The solution is complemented by an indigenously developed mobile application that enables simple parental control and monitoring.



SUPERVISOR

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GROUP MEMBERS

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OBJECTIVES

- ✓ Monitor internet traffic in real time.
- ✓ Identify inappropriate content for Kids.
- ✓ Detect Inappropriate & addictive content using AI.
- ✓ Block Inappropriate Content and Redirect Kids to Safe Content.
- ✓ Develop a mobile app for parental monitoring.
- ✓ Enable Kid Mode ON/OFF remotely.
- ✓ Provide network-level protection for all connected devices.



TOOLS & TECHNOLOGIES

HARDWARE



Raspberry Pi 5



Ethernet Cable



64GB MicroSD



Power Supply

SOFTWARE



Python



Scapy



MITMProv



CatBoost



RoBERTa



Flutter



Firebase



FEATURES



AI Based
Content Filtering



Real-time
Monitoring



Smart Blocking
& Redirection



Kid Mode
ON/OFF



Mobile App
Monitoring



Network-Level
Protection



APPLICATIONS



Home



Internet Service
Providers
(Nayatel, PTCL)



Educational
Institutions



Child-Centric
Organizations



Mobile Learning
Environments etc.

RESULTS

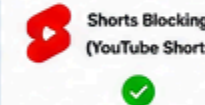
Customized Homepage

Redirection to Safe Content

Blocked Inappropriate Content



URL Monitoring
in Real Time



Shorts Blocking
(YouTube Shorts)



Platform Blocking
(TikTok, etc.)



Safe Content
Redirection



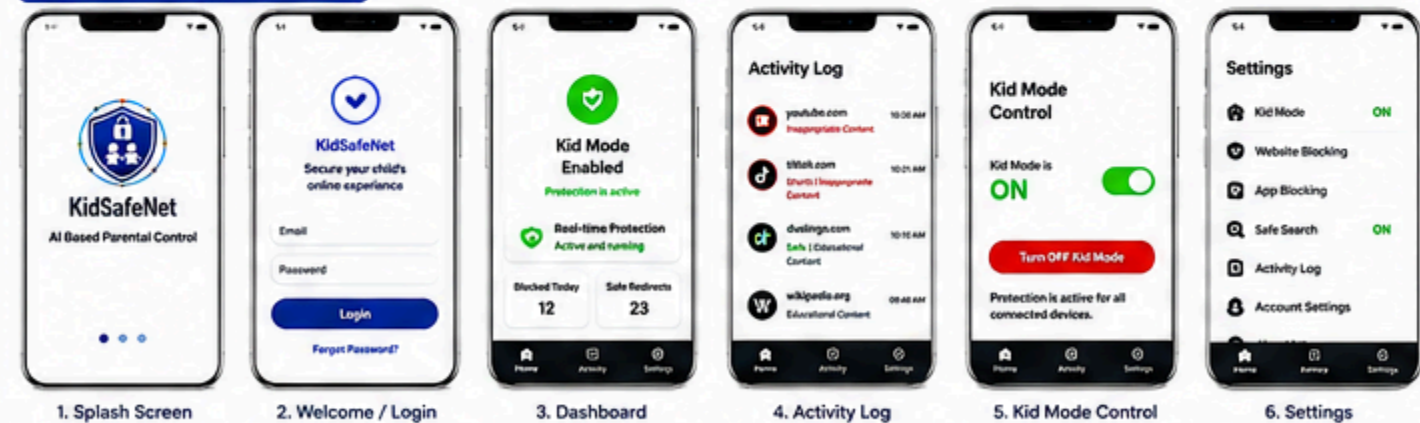
Kid Mode
ON / OFF



Activity Logging
in Mobile App



MOBILE APPLICATION



SDG 16



SCAN ME



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